

```
1 package ques2EcommercePlatformSearchFunction;
2
3 public class Product {
4
5     int productId;
6     String productName;
7     String category;
8
9     public Product(int productId, String productName, String category) {
10         this.productId = productId;
11         this.productName = productName;
12         this.category = category;
13     }
14
15     @Override
16     public String toString() {
17         return "ID: " + productId +
18             ", Name: " + productName +
19             ", Category: " + category;
20     }
21
22 }
23
```

```
1 package ques2EcommercePlatformSearchFunction;
2 import java.util.Arrays;
3 import java.util.Comparator;
4
5 public class SearchOperations {
6
7     // Linear Search
8     public static Product linearSearch(Product[] products, int id) {
9
10         for (Product product : products) {
11             if (product.productId == id) {
12                 return product;
13             }
14         }
15         return null;
16     }
17
18     // Binary Search
19     public static Product binarySearch(Product[] products, int id) {
20
21         int low = 0;
22         int high = products.length - 1;
23
24         while (low <= high) {
25
26             int mid = (low + high) / 2;
27
28             if (products[mid].productId == id) {
29                 return products[mid];
30             }
31             else if (products[mid].productId < id) {
32                 low = mid + 1;
33             }
34             else {
35                 high = mid - 1;
36             }
37         }
38     }
39 }
```

```
38
39     return null;
40 }
41
42 // Sort Products by ID
43 public static void sortProducts(Product[] products) {
44
45     Arrays.sort(products, Comparator.comparingInt(p -> p.productId));
46 }
47 }
```

```
1 package ques2EcommercePlatformSearchFunction;  
2 public class SearchTest {  
3  
4     public static void main(String[] args) {  
5  
6         Product[] products = {  
7  
8             new Product(103, "Mouse", "Electronics"),  
9             new Product(101, "Laptop", "Electronics"),  
10            new Product(104, "Keyboard", "Electronics"),  
11            new Product(102, "Shoes", "Fashion")  
12        };  
13  
14        // Linear Search  
15        System.out.println("Linear Search:");  
16        Product result1 = SearchOperations.linearSearch(products, 104);  
17  
18        if (result1 != null)  
19            System.out.println(result1);  
20  
21        // Binary Search  
22        SearchOperations.sortProducts(products);  
23  
24        System.out.println("\nBinary Search:");  
25        Product result2 = SearchOperations.binarySearch(products, 104);  
26  
27        if (result2 != null)  
28            System.out.println(result2);  
29    }  
30 }
```

Linear Search:

ID: 104, Name: Keyboard, Category: Electronics

Binary Search:

ID: 104, Name: Keyboard, Category: Electronics